RC-156 & RC-157 Execution Summary

24D-DMF Framework — Research Cycles 156 and 157

Date: 2026-07-11

Framework Version: v8.4.6

RC-156: The Interaction Engine — Unified Vertex Dynamics

Status: PARTIAL — 4/5 Criteria Pass

Criterion	Test	Result
C1c	Physical processes dominate	**PASS**
C2c	Orange couples universally	**PASS**
C3c	Yellow self-interactions exist	**PASS**
C4c	Green has minimal shattering	**FAIL**
C5c	Blue creates largest jumps	**PASS**

Key Findings

- The vertex Hamiltonian H = |ΔA | × |ΔA | measures shattering strength
- Physical processes (Orange/Blue mediated) dominate total shattering
- Orange (Gravity) couples to all 5 colors universally
- Yellow (QCD) self-interactions and split vertices confirmed
- Blue (Weak) creates the largest shattering jumps
- Green (QED) does NOT show minimal shattering — an anomaly requiring follow-up

Next Step

RC-157: Investigate whether shattering strength encodes energy-scale dependence (running couplings).

RC-157: The Running Engine — Couplings from Shattering Scale

Status: RUNNING ENGINE CONFIRMED

Criterion	Test	Result
C1	Amplitude correlates with energy scale	**PASS**
C2	Yellow (QCD) runs strong at IR	**FAIL**
C3	Green (QED) runs weak at IR (screening)	**PASS**
C4	Blue (Weak) has SSB threshold	**PASS**

Table 2 – continued

Criterion	Test	Result
C5	Beta function is geometric ($R^2 > 0.7$)	**FAIL**

Pass Condition: C1 AND (C2 OR C3) AND (C4 OR C5) = **TRUE**

Engines Used (v8.4.6 Math Reference)

- **H** : Unperturbed Hamiltonian (P + P † + 3(P + P †)) in 22D engine space
- **V**: Rank-2 orbit-swap perturbation (A B + B A)
- **K** : Skew-symmetric coupling operator from Gram =3 eigenspace
- **H_unified**: H + K + V, with = 3/√11, = (1+2/23)/(46√11)
- **Energy scale proxy**: Eigenvalue index in unified spectrum (0 = deep IR, 21 = UV)

Key Findings

Finding	Detail
Scale dependence (C1)	Spearman = 0.590 (p = 0.004) —shattering amplitude increases with energy
QED screening (C3)	Green absent from IR, present only in UV —photon decouples at low energy
Weak SSB threshold (C4)	Blue amplitude jumps 750× (0.0016 → 1.22) at Higgs energy scale
QCD reversed (C2)	Yellow runs WEAKER at IR, stronger at UV —opposite of asymptotic freedom
Beta function (C5)	Discrete, not continuous; best fit $R^2 = 0.37$ (log-periodic)

Honest Limitations

1. **Hybrid construction**: Color mapping uses Floquet projection pipeline on 22D eigenstates embedded in 24D
2. **Discrete sampling**: Only 22 eigenstates; running is sampled, not continuous
3. **Degenerate pairs**: Each H pair has one high-amp, one low-amp state — creates beta-function noise
4. **QCD anomaly**: Reversed running suggests “color = orbit mixing” may need revision for strong force

Next Step

RC-158: Build the 4-point vertex model (4-gluon, 4-photon).

Framework Context

RC	Contribution	Relevance to RC-157
RC-156	Vertex Hamiltonian H	Provides shattering strength measure
RC-155c	Gauge dynamics, golden ratio	Provides base amplitude and color mapping
RC-154b	Shattering confirmed	Provides shattering amplitude data
RC-152	Mass from wavelength	Provides mass-energy relation
RC-150b	Tunnel structure	Provides polar axis (UV fixed point)

End of Summary